Experiment 26: Machine Translation using Transformers in Python

1. Aim

To write a Python program to implement machine translation from English to French using the Hugging Face Transformers library.

2. Procedure

1. Import `pipeline` from the `transformers` library.
2. Initialize the translation pipeline using `pipeline('translation_en_to_fr')`.
3. Define a sample English sentence to translate.
4. Pass the English text to the translator pipeline.
5. Display the English input and the translated French text.

3. Program

```
from transformers import pipeline

translator = pipeline("translation_en_to_fr")
text = "Python is easy to learn and very useful."

output = translator(text)
print("English:", text)
print("French:", output[0]['translation_text'])
```

4. Output

```
English: Python is easy to learn and very useful.
French: Python est facile à apprendre et très utile.
```

5. Result

Thus, the Python program was successfully executed to translate English text to French using the Hugging Face `transformers` library.